

NIELIT CCC • UNIT 06

Introduction to Internet and WWW

Curriculum Hours: 7h (3h Theory + 4h Lab) | Topics: 8 | Language: English

UNIT OVERVIEW & PEDAGOGICAL SCOPE

The Internet is the global interconnected network of computer networks, while the World Wide Web (WWW) is the vast information space of hyperlinked multimedia documents accessed across the Internet. This unit covers foundational networking concepts: network scales (PAN, LAN, MAN, WAN), network topologies (Star, Bus, Ring, Mesh, Tree), Internet history (ARPANET, Vint Cerf, VSNL in India), URL structure, IP addressing (IPv4 vs IPv6), MAC physical addresses, IMEI numbers, standard protocols and port numbers, broadband access modes, web browsers, and search engines.

WHAT YOU WILL MASTER IN THIS UNIT

- Network classifications by geographical scale: PAN (<10m, Bluetooth), LAN (Office/Lab), MAN (City), and WAN (Global Internet).
- Network topologies: Star (central hub/switch), Bus (single backbone cable), Ring (token loop), Mesh (point-to-point redundant links), and Tree.
- The historical genesis of the Internet: ARPANET (1969), Vinton Cerf & Bob Kahn, and VSNL public launch in India (15 August 1995).
- The World Wide Web (Tim Berners-Lee, 1989 at CERN), the difference between Internet and WWW, and URL anatomy (Protocol, Domain, Port, Path, Query).
- Device identification: IPv4 (32-bit dotted-decimal) versus IPv6 (128-bit hexadecimal colons), 48-bit MAC physical address, and 15-digit IMEI (*#06#).
- Core Internet protocols and standard port numbers: TCP/IP, HTTP (80), HTTPS (443), FTP (21), DNS (53), SMTP (25), POP3 (110), IMAP (143).
- Internet connection modes: FTTH optical fiber, Wi-Fi 802.11, mobile Wi-Fi Hotspots, and USB/Bluetooth tethering.
- Web Browsers (Mosaic 1993, Chrome, Firefox, Lynx text-only) and Search Engines (Google, DuckDuckGo, Web Crawlers, Indexing, SERP).

TOPIC 6.1: BASICS OF COMPUTER NETWORKS

A computer network is an interconnected group of autonomous digital devices (computers, servers, printers) linked via physical cables or wireless signals to share data, files, software, and hardware resources.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Stand-alone computers cannot communicate or share peripherals. Networking enables instantaneous global file transfers, centralized database access, hardware cost reduction, and internet access.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Network Interface Cards (NICs) convert computer data into physical signals (electrical voltages, light pulses, radio waves). The signals travel across media to receiving devices that reconstruct the data packets.

PRIMARY BENEFITS OF COMPUTER NETWORKING

- Hardware Resource Sharing: Multiple office computers share a single high-speed laser printer, scanner, or central storage server.
- Centralized Data Management: Company databases reside on secure central servers rather than scattered across individual desktops.
- Rapid Communication: Instant transmission of emails, messages, video conferences, and file synchronization.

- Cost Efficiency: Eliminates purchasing redundant individual software licenses and peripheral devices.

CLASSIFICATION OF NETWORKS BY GEOGRAPHICAL SPAN

Network Type	Geographical Coverage Range	Common Physical Technologies	Real-World Deployment Example
PAN (Personal Area Network)	Under 10 meters (Personal workspace)	Bluetooth (IEEE 802.15), USB, Zigbee	Connecting wireless earbuds, smartwatches, or wireless mice to a phone
LAN (Local Area Network)	Up to 1 kilometer (Single room, building, lab)	Ethernet twisted-pair cables (RJ-45), Wi-Fi (802.11)	A university computer lab, hospital building, or corporate office network
MAN (Metropolitan Area Network)	10 to 50 kilometers (Entire town or city)	Fiber-optic backbone, high-speed microwave links	City-wide cable television networks or municipal police camera networks
WAN (Wide Area Network)	Country, continent, or entire globe	Submarine optical cables, communication satellites	The global Internet, national banking networks (RBI CBS, ATM grid)

In an educational computer lab with 40 PCs: Instead of buying 40 separate printers, all 40 PCs are connected to a single Local Area Network (LAN). Any student can send their assignment to the single shared network printer in the room.

- PAN covers personal workspace distances under 10 meters (Bluetooth).
- LAN covers a single room, school lab, or building.
- MAN covers a city or metropolitan area.
- WAN covers entire countries or continents; the Internet is the largest WAN in existence.

QUICK TOPIC RECAP

Networks connect devices to share resources: PAN (<10m, Bluetooth), LAN (building/lab), MAN (city), WAN (global internet). The Internet is the largest WAN.

TOPIC 6.2: NETWORK TOPOLOGIES: STAR, BUS, RING, MESH, TREE

Network topology refers to the geometric physical or logical arrangement of nodes, workstations, and connecting cables in a computer network.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Topology choice directly impacts network installation costs, data transfer reliability, ease of troubleshooting, and fault tolerance against cable breaks.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Different topologies route data according to different geometric patterns: centralized hubs, shared continuous cables, closed circular loops, or redundant point-to-point meshes.

TOPOLOGY CALCULATIONS & CONCEPTS

- Star Topology Dominance: Modern Ethernet LANs exclusively use Star topology powered by intelligent central network switches.
- Bus Terminators: Bus networks require electrical terminators at both ends of the main backbone cable to prevent signal reflection bounces.
- Mesh Formula: In a full mesh network with n computers, the total number of physical cables required is calculated as: $\frac{n(n-1)}{2}$ (e.g. 6 computers require 15 cables).

COMPARISON OF THE 5 MAJOR NETWORK TOPOLOGIES

Topology Name	Physical Architecture / Layout	Key Advantages	Critical Disadvantages / Vulnerabilities
Star Topology	Every node connects individually to a central Hub or Switch	Easy to add nodes; failure of one cable does not affect others	If the central Hub/Switch fails, the entire network collapses
Bus Topology	All nodes connect to a single shared backbone cable with terminators	Inexpensive, minimal cabling required; easy to set up for small labs	Backbone cable break halts entire network; heavy data collisions
Ring Topology	Nodes connected in a closed circle; data passes as a token in 1 direction	No packet collisions; predictable equal access time for all nodes	Break in the physical ring cable breaks the entire network token loop
Mesh Topology	Every single node connects directly to every other node (Point-to-point)	Maximum fault tolerance; failure of one link does not stop traffic	Extremely expensive; complex cabling requiring $n(n-1)/2$ physical links
Tree Topology	Hierarchical tree structure combining multiple star networks along a bus	Highly scalable; easily segments large corporate departments	Heavily dependent on the main root trunk cable backbone

In a modern school computer lab: Every desktop is plugged with an Ethernet cable into a central 24-port network switch on the wall (Star Topology). If a student accidentally unplugs computer 5, the remaining 39 computers continue browsing the internet with zero interruption.

- Star topology is the most widely used topology in modern LANs.
- In Star topology, all computers connect to a central Hub or Switch.
- In Bus topology, all computers share a single central backbone cable.
- Total physical links in a fully connected Mesh topology with `n` nodes: $n(n - 1) / 2$.
- Ring topology uses unidirectional token passing.

QUICK TOPIC RECAP

Topologies: Star (central hub/switch, most popular), Bus (single cable with terminators), Ring (circular token loop), Mesh (point-to-point, max redundancy), Tree (hierarchical).

TOPIC 6.3: INTERNET ARCHITECTURE, HISTORY, AND APPLICATIONS

The Internet is a global system of interconnected computer networks that communicate using the standard TCP/IP (Transmission Control Protocol / Internet Protocol) suite.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

The Internet is the foundational communications backbone supporting international commerce, financial transactions, news broadcasting, governance, and research.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Computers connect to local Internet Service Providers (ISPs), which connect to regional Tier-2 networks, which route traffic across ultra-high-speed transcontinental Tier-1 submarine optical fiber cables spanning the oceans.

HISTORICAL MILESTONES OF THE INTERNET

- ARPANET (1969): Advanced Research Projects Agency Network, funded by the US Department of Defense; the world's first operational packet-switching network connecting four universities.
- Fathers of the Internet: Vinton Gray Cerf and Robert Elliot Kahn (designed the TCP/IP protocol suite in the 1970s).
- Public Launch in India (15 August 1995): Videsh Sanchar Nigam Limited (VSNL) officially launched public commercial Internet access in India under Chairman Brijendra K. Syngal.
- Internet Service Provider (ISP): Commercial entities (BSNL, Jio, Airtel) providing internet connectivity to homes, colleges, and enterprises.

MAJOR PRACTICAL APPLICATIONS OF THE INTERNET

Application Domain	Core Online Services	Everyday Citizen Impact
Communication	Email, Instant Messaging, Video Calling (VoIP)	Instant global communication at virtually zero cost
Digital Banking & Finance	UPI, NEFT, RTGS, IMPS, Net Banking	24x7 cashless fund transfers and automated bill payments
E-Commerce	Online Shopping, Ticket Booking (IRCTC), Food Delivery	Buying goods and reserving train tickets directly from home
E-Governance	DigiLocker, Passport Seva, Income Tax e-filing, UMANG	Direct, transparent citizen government services without office queues

When you book a railway ticket on IRCTC: Your phone connects via your ISP (Jio/Airtel) to IRCTC servers in New Delhi, verifies seat availability via central railway databases, routes payment through your bank gateway, and issues an SMS/QR ticket in seconds.

- The precursor to the modern Internet was ARPANET, established in 1969.
- Vinton Cerf and Bob Kahn are recognized as the Fathers of the Internet.
- Public Internet was officially launched in India on 15 August 1995 by VSNL.
- The fundamental communication protocol suite of the Internet is TCP/IP.

QUICK TOPIC RECAP

The Internet originated from ARPANET (1969). Vint Cerf and Bob Kahn are Fathers of the Internet. VSNL launched public internet in India on 15 August 1995. Built on TCP/IP.

TOPIC 6.4: WORLD WIDE WEB (WWW) AND URL ANATOMY

The World Wide Web (WWW or Web) is a global collection of interlinked multimedia hypertext documents (web pages) accessed via the Internet using web browsers.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

The Web is often confused with the Internet. Understanding their distinction and mastering URL anatomy is critical for web literacy and internet security.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

The Internet is the physical networking infrastructure (cables, routers, computers), while the WWW is an application service that runs ON TOP OF the Internet using Hypertext Transfer Protocol (HTTP).

CRITICAL DISTINCTION: INTERNET VS. WORLD WIDE WEB

- Internet = Physical Highway: The physical infrastructure of wires, routers, fiber optics, satellites, and computers.
- World Wide Web = Traffic on Highway: The service of hyperlinked web pages, text, and images created by Sir Tim Berners-Lee in 1989 at CERN (Switzerland).
- W3C (World Wide Web Consortium): The international standards organization founded by Tim Berners-Lee that develops open web standards (HTML, CSS).

ANATOMY OF A UNIFORM RESOURCE LOCATOR (URL)

URL Component	Example in URL: https://www.nielit.gov.in:443/student/index.html	Function & Technical Meaning
Protocol / Scheme	https://	Hypertext Transfer Protocol Secure; encrypts transmission using SSL/TLS
Subdomain	www	World Wide Web host designation or specific service name
Domain Name	nielit.gov.in	Unique human-readable registered address of the web server

Port Number	:443 (Default for HTTPS; 80 for HTTP)	Designates the specific network service communication port
Directory Path	/student/	Sub-folder directory path on the server where the web page is stored
Resource File	index.html	Specific target web document file rendered by the browser

Top-Level Domains (TLDs) indicate the category or geographical origin of a website:

- `.com`: Commercial business entities (e.g. google.com)
- `.gov` / `.gov.in`: Official government entities (e.g. nielit.gov.in)
- `.edu` / `.ac.in`: Accredited educational colleges and universities
- `.org`: Non-profit organizations
- `.in`: Country-code top-level domain (ccTLD) for India.

- Sir Tim Berners-Lee invented the World Wide Web in 1989 at CERN.
- The Internet and the World Wide Web are NOT the same thing; WWW runs on top of the Internet.
- URL stands for Uniform Resource Locator.
- HTTPS uses port 443; HTTP uses port 80.
- `.in` is the country code top-level domain for India; `.gov` represents government websites.

QUICK TOPIC RECAP

WWW was invented by Tim Berners-Lee (1989 at CERN). Internet is the physical network; WWW is the web pages on it. URL = Protocol + Domain + Port + Path + File.

TOPIC 6.5: IP ADDRESSING, MAC ADDRESS, AND IMEI

Digital devices rely on three distinct numbering standards for identification: IP Addresses (logical network routing), MAC Addresses (physical hardware identification), and IMEI Numbers (mobile cellular hardware identification).

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Network routers cannot deliver web pages or emails without precise numeric addresses, and cellular networks cannot authenticate phones without hardware IDs.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

An IP address identifies a device's current logical position on a network. A MAC address is permanently burned into the device's NIC silicon. An IMEI identifies a phone to cellular telecom towers.

IMEI (INTERNATIONAL MOBILE EQUIPMENT IDENTITY)

- Structure: Exactly 15 decimal digits unique to every physical SIM slot on mobile phones.
- Check Code: Dialing `\*#06#` on any mobile phone keypad immediately displays its IMEI numbers on screen.
- Purpose: Telecom operators use IMEI numbers to blacklist and block stolen phones from accessing cellular networks.

COMPARISON: IPv4 VS. IPv6 VS. MAC ADDRESS

Attribute	IPv4 (IP Version 4)	IPv6 (IP Version 6)	MAC Address (Media Access Control)
Address Length	32 Bits (4 Bytes)	128 Bits (16 Bytes)	48 Bits (6 Bytes)
Format & Notation	Dotted Decimal notation (e.g. 192.168.1.1)	Hexadecimal separated by Colons (e.g. 2001:db8::1)	Hexadecimal separated by colons or hyphens (00:1A:2B:3C:4D:5E)
Number of Parts	4 Octets (each 0 to 255)	8 Hexadecimal groups (each 16 bits)	6 Hexadecimal byte octets (First 3 = Vendor ID, Last 3 = Device ID)
Nature & Persistence	Logical address; changes when connecting to different networks	Logical address; massive address space (2^{128} addresses)	Physical hardware address; permanently hardcoded into the network card

Total Address Space	~4.3 Billion (2^{32} addresses; exhausted)	~340 Undecillion (Virtually unlimited)	2^{48} unique physical network interface cards
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When you buy a smartphone: It has a permanent 15-digit IMEI printed on the box. When it connects to home Wi-Fi, its Wi-Fi chip uses its 48-bit MAC address. The Wi-Fi router assigns it a temporary 32-bit IPv4 address (like `192.168.1.15`) to browse the internet.

- IPv4 addresses are 32 bits (4 bytes) long and written in dotted-decimal format.
- IPv6 addresses are 128 bits (16 bytes) long and written in hexadecimal colon format.
- MAC address is 48 bits (6 bytes) long and represents the physical hardware address of the NIC.
- IMEI stands for International Mobile Equipment Identity and contains exactly 15 digits.
- The universal USSD code to check a phone's IMEI number is `\*#06#`.

QUICK TOPIC RECAP

IPv4 = 32-bit (dots), IPv6 = 128-bit (colons), MAC = 48-bit physical hardware address. IMEI = 15-digit mobile identity checked via *#06#.

TOPIC 6.6: NETWORK PROTOCOLS AND STANDARD PORT NUMBERS

A network protocol is a formal, standardized set of rules and conventions governing how computers format, transmit, verify, and receive data across networks.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Computers built by different manufacturers using different operating systems could never communicate without standardized protocols and port numbers.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Protocols operate at different layers: TCP breaks messages into numbered packets and guarantees delivery; IP addresses packets; and application protocols (HTTP, FTP, SMTP) deliver specific services through dedicated communication port numbers.

CORE INTERNET PROTOCOLS & DEFAULT PORT NUMBERS

Protocol Acronym	Full Protocol Name	Default Port Number	Primary Functional Purpose in Networking
HTTP	Hypertext Transfer Protocol	Port 80	Transmits unencrypted web pages across the World Wide Web
HTTPS	Hypertext Transfer Protocol Secure	Port 443	Transmits encrypted, secure web pages using SSL/TLS encryption
FTP	File Transfer Protocol	Port 20 / 21	Transfers, uploads, and downloads files between client and server
DNS	Domain Name System	Port 53	Translates human-readable domain names (nielit.gov.in) into IP addresses
SMTP	Simple Mail Transfer Protocol	Port 25	Sends and relays outgoing emails from email clients to mail servers
POP3	Post Office Protocol version 3	Port 110	Downloads incoming emails from mail server to local device
IMAP	Internet Message Access Protocol	Port 143	Synchronizes incoming emails across multiple devices in real time
SSH	Secure Shell	Port 22	Secure encrypted remote command-line login and server administration
DHCP	Dynamic Host Configuration Protocol	Port 67 / 68	Automatically assigns IP addresses to devices connecting to a network

When you type `https://nielit.gov.in` into your browser: The browser contacts a DNS server (Port 53) to find the server's numeric IP address. It then establishes a secure encrypted HTTPS connection (Port 443) using SSL/TLS certificates to load the official page safely.

- HTTP uses Port 80; HTTPS uses Port 443.
- DNS (Port 53) translates domain names into numerical IP addresses.
- SMTP (Port 25) is used for SENDING outgoing emails.
- POP3 (Port 110) and IMAP (Port 143) are used for RECEIVING incoming emails.
- FTP uses Port 20 and 21 for file transfers.

QUICK TOPIC RECAP

Protocols govern communication. Key ports: HTTP = 80, HTTPS = 443, FTP = 21, DNS = 53, SMTP = 25 (send mail), POP3 = 110 (receive mail), IMAP = 143 (sync mail).

TOPIC 6.7: INTERNET ACCESS MODES: FTTH, WI-FI, HOTSPOT, TETHERING

Internet access modes are the physical and wireless transmission channels through which end-user devices connect to an Internet Service Provider (ISP).

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Choosing the correct access mode balances bandwidth speed, installation cost, mobility, and signal reliability.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

Access modes range from fixed physical guided media (optical fiber carrying light pulses) to unguided wireless radio frequency bands (Wi-Fi and cellular modems).

OVERVIEW OF INTERNET ACCESS MODES

Access Mode	Transmission Medium	Speed & Coverage	Typical Practical Use Case
FTTH (Fiber to the Home)	Glass / Silica Optical Fiber cables	Ultra-fast (100 Mbps to 1 Gbps+); unlimited data	Broadband internet for homes, offices, schools, and computer labs
Wi-Fi (Wireless Fidelity)	Radio Waves (2.4 GHz & 5 GHz bands, IEEE 802.11)	High speed (50 to 500+ Mbps); range ~30-50 meters	Wireless local networking for laptops, phones, and smart TVs
Mobile Wi-Fi Hotspot	4G/5G Cellular data converted to Wi-Fi	Moderate to High speed; range ~10 meters around phone	Sharing a smartphone's internet connection with a nearby laptop
USB Tethering	Physical USB cable connecting phone to PC	High speed; charges phone simultaneously; zero Wi-Fi radiation	Connecting a desktop PC that lacks a Wi-Fi card to mobile internet
Bluetooth Tethering	Low-power 2.4 GHz Bluetooth radio signals	Lower speed (<2 Mbps); low battery drain; range ~10 meters	Sharing low-bandwidth mobile internet between handheld devices

If a desktop PC in an exam room loses broadband internet because a fiber cable was cut outside: The proctor connects their smartphone to the PC using a USB charging cable and enables 'USB Tethering' in phone settings. The desktop instantly resumes internet access through the phone's mobile data.

- FTTH stands for Fiber to the Home.
- Wi-Fi operates on 2.4 GHz and 5 GHz radio frequency bands (IEEE 802.11 standards).
- USB Tethering allows sharing a smartphone's mobile data via a physical USB cable.
- Mobile Hotspot allows sharing mobile data wirelessly with multiple devices over Wi-Fi.

QUICK TOPIC RECAP

Access modes: FTTH (fastest fiber), Wi-Fi (wireless local network), Mobile Hotspot (wireless phone sharing), USB Tethering (wired phone sharing), Bluetooth Tethering (low-power sharing).

TOPIC 6.8: WEB BROWSERS AND SEARCH ENGINES

A **Web Browser** is an application software program used to locate, retrieve, and render web pages from the WWW. A **Search Engine** is a specialized web service that indexes billions of web pages and helps users discover relevant information by searching keywords.

2. WHY IS IT IMPORTANT IN INDUSTRY & EXAMS?

Without a web browser, users cannot view HTML/CSS web documents. Without a search engine, navigating the billions of websites on the internet would be impossible.

3. HOW DOES IT WORK? (MECHANISM & DATA FLOW)

A user types a search query into a search engine (like Google). The search engine queries its massive database of indexed pages (gathered by automated **Web Crawlers / Spiders**) and displays a Search Engine Results Page (SERP). The user clicks a link, and their web browser connects to the target server to render the page.

ESSENTIAL BROWSER SHORTCUTS & HISTORY

- Mosaic (1993): The world's first widely popular graphical web browser (developed by Marc Andreessen at NCSA).
- Lynx: A popular text-only web browser that runs inside terminal command-line consoles without rendering images.
- Add Bookmark: Press `Ctrl + D` to save the active web page to your favorites bar.
- View Browsing History: Press `Ctrl + H` to view previously visited websites.
- View Downloads: Press `Ctrl + J` to inspect downloaded files.
- New Incognito / Private Window: Press `Ctrl + Shift + N` in Chrome/Edge (or `Ctrl + Shift + P` in Firefox) to browse without saving history or cookies.
- Save Current Web Page: Press `Ctrl + S` to save an offline copy of the active web page.

WEB BROWSERS VS. SEARCH ENGINES COMPARISON

Parameter	Web Browser (Application Software)	Search Engine (Online Web Service)
Core Nature	Software program installed locally on your computer or phone	Web-based online software database hosted on remote server clusters
Primary Purpose	Renders HTML code, executes JavaScript, and displays web pages	Discovers, catalogs, indexes, and ranks web page addresses by topic keywords
Installation Required	Yes: Must be installed (or pre-installed with the OS)	No: Accessed by visiting its website address inside any browser
Prominent Examples	Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera, Lynx	Google, Microsoft Bing, Yahoo Search, DuckDuckGo (privacy-focused)

Google Chrome is the Web Browser software installed on your laptop. When you open Chrome and type 'NIELIT CCC exam syllabus', Google is the Search Engine that processes your words, scans its index, and displays links to official NIELIT web pages.

- Google Chrome is a Web Browser; Google is a Search Engine.
- Mosaic was the first widely used graphical web browser (1993).
- Lynx is a famous text-only web browser.
- Shortcut to Bookmark a web page: `Ctrl + D`.
- Shortcut to view Browsing History: `Ctrl + H` | Downloads: `Ctrl + J`.
- Search engines use automated software programs called 'Web Crawlers' or 'Spiders' to index web pages.

QUICK TOPIC RECAP

Browser = software to view web pages (Chrome, Firefox, Lynx text-only). Search Engine = online service to search the web (Google, DuckDuckGo). Bookmarks = Ctrl+D, History = Ctrl+H.

TOPIC undefined:

- Networks connect autonomous devices to share data and peripherals: PAN (<10m, Bluetooth), LAN (building/lab), MAN (city), WAN (global Internet).
- Network Topologies: Star (central switch, most reliable/popular), Bus (single cable with terminators), Ring (token loop), Mesh (point-to-point, $n(n-1)/2$ links), Tree (hierarchical).
- The Internet originated from ARPANET (1969). Vinton Cerf and Bob Kahn are the Fathers of the Internet. VSNL launched public internet in India on 15 August 1995.
- The World Wide Web (WWW) was invented by Sir Tim Berners-Lee in 1989 at CERN. Internet is the physical network; WWW is the multimedia service running on it.
- URL (Uniform Resource Locator) structure: Protocol (https) + Domain (nielit.gov.in) + Port (443) + Path + File.
- Device Identification: IPv4 is 32-bit (dotted-decimal, 4 bytes); IPv6 is 128-bit (hexadecimal colons, 16 bytes); MAC is 48-bit physical NIC address; IMEI is 15-digit mobile ID (*#06#).
- Core Protocols: HTTP (Port 80), HTTPS (Port 443), FTP (Port 20/21), DNS (Port 53), SMTP (Port 25 - send mail), POP3 (Port 110 - receive mail), IMAP (Port 143 - sync mail).
- Internet Access: FTTH (optical fiber broadband), Wi-Fi (802.11 2.4/5GHz), Mobile Hotspot, USB Tethering, Bluetooth Tethering.
- Web Browsers (Chrome, Firefox, Lynx text browser) render web pages. Search engines (Google, DuckDuckGo) index pages using Web Crawlers / Spiders.
- Browser Shortcuts: Bookmark = `Ctrl + D`, History = `Ctrl + H`, Downloads = `Ctrl + J`, Incognito = `Ctrl + Shift + N`.

KEY TERMS, ACRONYMS & OFFICIAL FULL FORMS

--- END OF UNIT 06 DETAILED STUDY BOOK • NIELIT CCC (80 HOURS) ---